



Computer Science Department
COMP233-Discrete Mathematics / Spring 2026
Section #4 — Exam#2 — Chapters 5,6
Time: 60 Minutes — Total Marks: 100

Student Name: _____

Student ID: _____

Please, use the answer sheet form ONLY!

Question 1 (18 Marks): Fill **one** correct answer for each branch in the table below.

1. Which A(an) _____ is a sequence of the form $a + ar + ar^2, \dots, ar^n, \dots$			
a) Arithmetic progression		b) Geometric progression	
c) Fibonacci		d) Floor function	
2. The difference of the sets A and B (i.e., $A - B$) is equal to _____.			
a) $\{x x \in A \vee x \in B\}$	b) $\{x x \in A \wedge x \notin B\}$	c) $\{x x \in A \vee x \notin B\}$	d) $\{x\}$
3. Which of the following set identities represents the correct translation of the Logical Absorption Law into Set Theory?			
a) $A \cup (A \cap B) = A$		b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	
c) $A - B = A \cap B^c$		d) $(A \cup B)^c = A^c \cap B^c$	
4. The set S of odd positive integers less than 7 can be expressed by using set builder notation as _____.			
a) $S = \{x \in \mathbb{R}^+ \mid x \text{ is odd and } x < 7\}$		b) $S = \{x \in \mathbb{Z}^+ \mid x \text{ is odd and } x < 7\}$	
c) $S = \{x \mid x \text{ is odd and } x < 7\}$		d) $S = \emptyset$	
5. For any two finite sets A and B, if the number of elements in the Power Set $\mathcal{P}(A)$ is 32, and $\mathcal{P}(B)$ has 8 elements, then the number of elements in the Cartesian Product $(A \times B)$ is:			
a) 15 Elements		b) 40 Elements	
c) 256 Elements		d) 24 Elements	
6. In the inductive step of proving that $1 \times 2 + 2 \times 3 + \dots + n(n + 1) = \frac{n(n+1)(n+2)}{3}$, after assuming the inductive hypothesis $P(k)$ is true, what specific term must be added to both sides to evaluate $P(k + 1)$?			
a) $k(k + 1)$		b) $(k + 1)(k + 2)$	
c) $k(k + 2)$		d) $(k + 1)^2$	
7. The following double sums $\sum_{i=0}^2 \sum_{j=0}^3 j$ is equal _____.			
a) 18	b) $6i$	c) $6j$	d) 12
8. A telescoping sum of the form: $\sum_{i=1}^n [i - (i + 1)]$ evaluates directly to:			
a) 0	b) $\frac{n(n+1)}{2}$	c) $-n$	d) $1 - n$
9. What is the most simplified form of this Boolean equation? $(A + \overline{B})(A + \overline{C})(B + C)$			
a) $A + \overline{C}$	b) $\overline{A} \cdot B + \overline{A} \cdot C + \overline{B} \cdot \overline{C}$	c) 1	d) $\overline{A} + \overline{B} + \overline{C}$

Question 2 (17 Marks):

- (a) The number of ancestors an artificial neural network node can trace back at level k is given by $A_k = 2^k$. (17 marks)

1. Compute the total number of ancestors at levels 1, 2, and 3.
2. Express the total number of ancestors across all levels from 1 to n using a single mathematical summation notation.
3. **PROVE** by expanding the terms and applying the geometric series formula that this total sum can be written in the closed form: $2^{n+1} - 2$.

Question 3 (35 Marks):

- (a) **PROVE** algebraically that for any integer $n \geq 1$:

$$\frac{(2n)!}{(2(n-1))!} = 4n^2 - 2n$$

Show your work step-by-step using the recursive definition of factorials, simplifying the expression completely without using any factorial notation. (10 marks)

- (b) Let the Universal set be the set of all integers $\mathbb{Z}$. Consider the following three parameterized subsets:

$$A = \{x \in \mathbb{Z} \mid 1 \leq x \leq 6\}$$

$$B = \{x \in A \mid x \text{ is an even number}\}$$

$$C = \{x \in \mathbb{Z} \mid x = 2k + 1 \text{ for some integer } k\}$$

Based on these definitions, analyze and determine the following, showing your logical or numerical justification for each step: (25 marks)

- 1) Determine the sets B and $A \cap C$ in roster form (e.g., $R = \{a, b, c, \dots\}$).
- 2) Find the Set Difference $A - B$ and its relationship to the complement set B^c .
- 3) Compute the Cartesian Product $(B \times (A \cap C))$ and state its cardinality.
- 4) Evaluate whether the sets B and C are **disjoint**. Justify using the definition.
- 5) Let $\mathcal{A} = \{B, (A - B)\}$. Evaluate whether the collection $\mathcal{A}$ forms a valid partition of the set A . Justify your answer formally using the concepts of mutual disjointness and set union.

Question 4 (30 Marks):

- (a) **PROVE** by Mathematical Induction that for all integers $n \geq 1$, the sum of the cubes of the first n positive integers satisfies the formula:

$$1^3 + 2^3 + 3^3 + \dots + n^3 = \left[\frac{n(n+1)}{2} \right]^2$$

Identify and clearly label your Basis Step, Inductive Hypothesis, and Inductive Step. (16 marks)

- (b) **PROVE** the following set identity algebraically by applying the standard laws of Set Identities:

$$(A \cup B) \cap (A \cup B^c) = A$$

State clearly the name of the specific law used at each step of your derivation. (14 marks)

Good Luck!



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Answer Sheet

Question 1 (18 Marks):

Fill your final answers clearly in the table below:

Question	1	2	3	4	5	6	7	8	9
Answer									

Question 2 (17 Marks):

Q2.(a.1) *Answer:*

Q2.(a.2) *Answer:*

Q2.(a.3) *Answer:*

Question 3 (35 Marks):**Q3.(a)** *Answer:*

Q3.(b) (1) *Answer:*

(2) *Answer:*

(3) *Answer:*

(4) *Answer:*

(5) *Answer:*

Question 4 (30 Marks):**Q4.(a)** *Answer:*

Q4.(b) *Answer:*

Extra Paper

Clearly label which question you are answering.



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Answer Key